Charging & Usage Guide

1. Charging Instruction

For delivery safety, the LiPo battery is not fully charged. Please fully charge the battery before the first operation. To do so, connect the battery to the battery connector before the first charge. The charging time is approximately one hour.

2. Battery Usage Guideline

1) **Charging**: Please charge the battery using the provided charger when the voltage drops below 6.4V. This helps maintain stable operation of the robot.
2) **Long-Term Storage**: If the robot will not be used for an extended period, fully charge the battery beforehand. Switch the expansion board to “OFF” and disconnect the battery wires.
3) **Storage Environment**: Store the battery in a cool, dry place to prevent damage caused by overheating or moisture, which can shorten battery life.
4) **Handling Precautions**: Do not hit, throw, or step on the battery.
5) **Usage Conditions**: Do not use the battery in environments with strong electrostatic or magnetic fields, as this may damage the battery's safety protection circuitry.

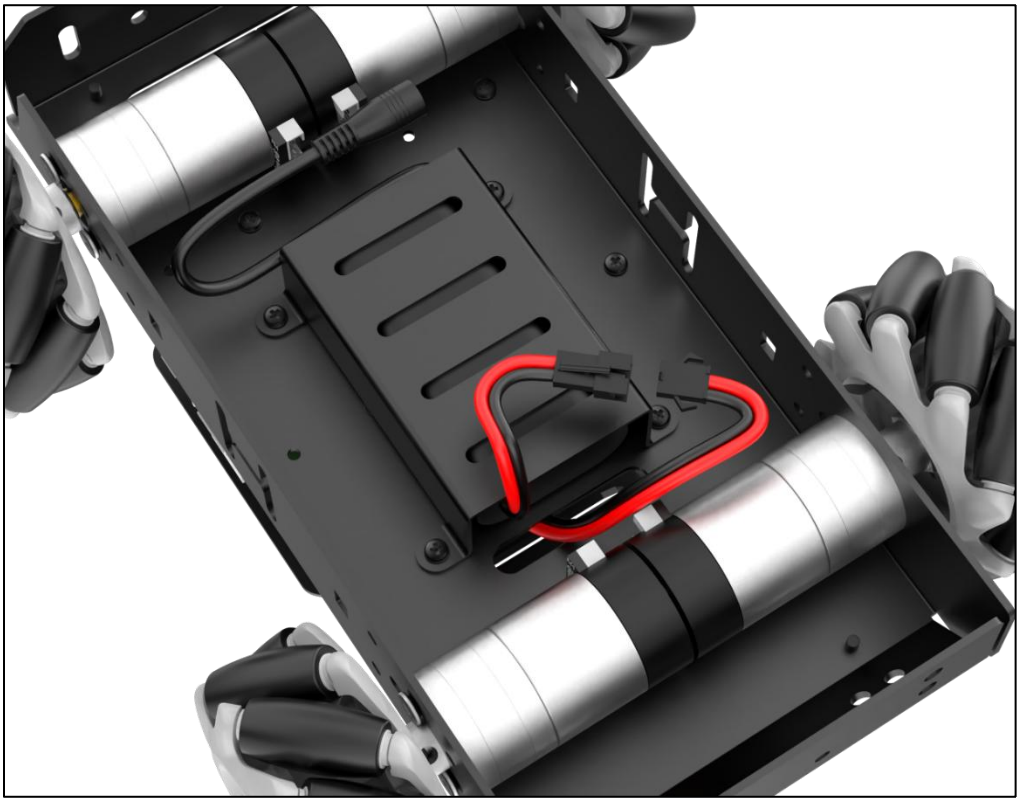
Warning: Please follow these guidelines strictly. Our company is not responsible for any product damage, economic loss, or accidents resulting from improper use.

3. Charging Method

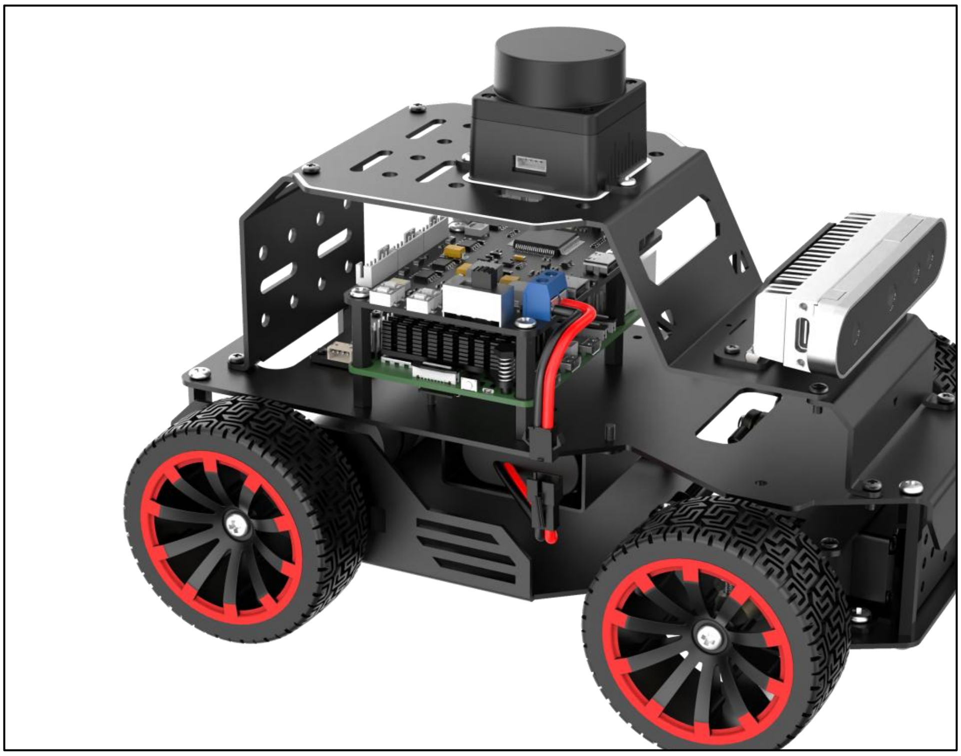
Charging steps are as follows:

1) Turn OFF the robot.
2) Connect the battery by matching the connectors: red to red, and black to

black.



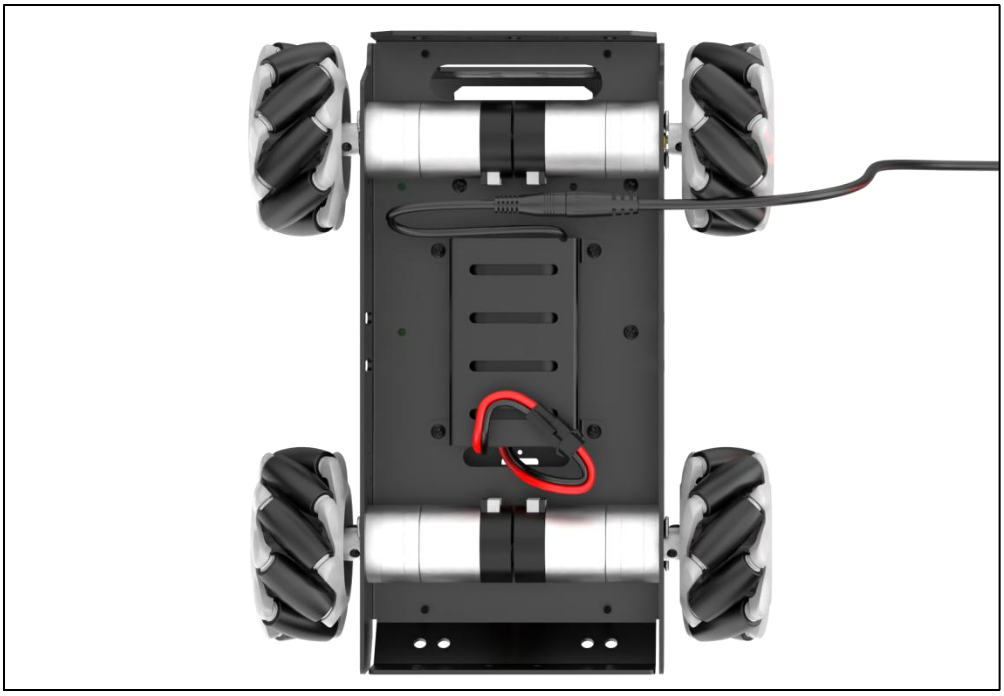
Battery Connector Location Diagram – Mecanum Wheel Chassis



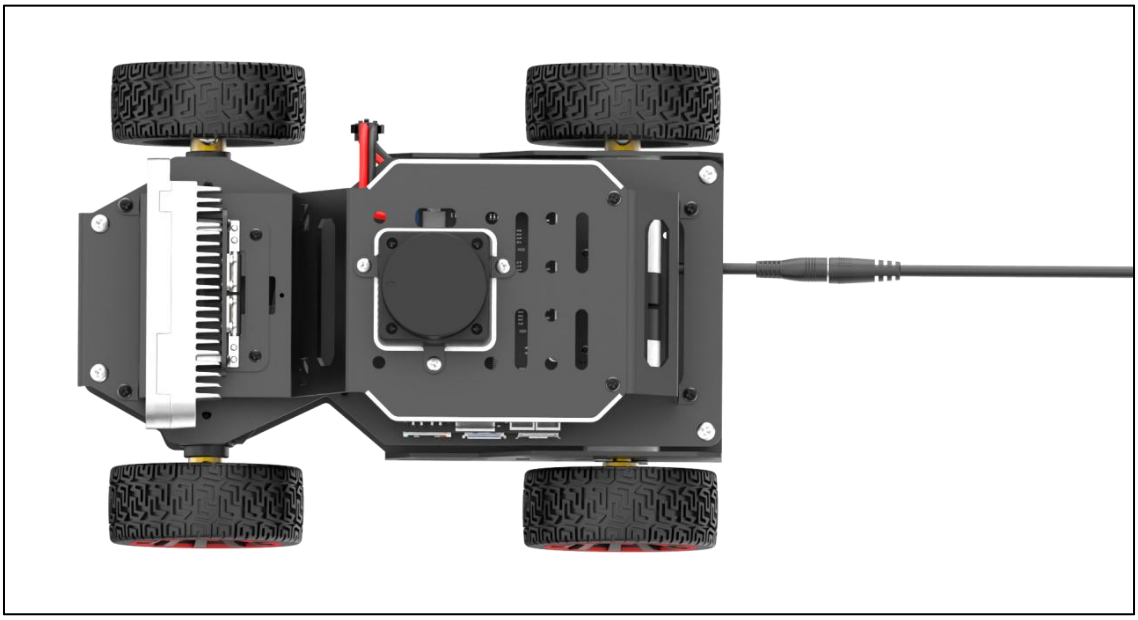
Battery Connector Location Diagram – Ackerman wheel chassis

3) Remove the lithium battery's DC plug from the side or bottom of the device,

then insert the charger connector to begin charging.



Charging port location diagram for Mecanum wheel chassis



Charging port location diagram for Ackerman wheel chassis

4) Check the charging status using the indicator light on the charger: red indicates charging, while green means the battery is fully charged.

Note: Please unplug the charger in time after charging is complete to avoid overcharging the battery.

4. Note for Device Usage

1) Do not place the robot on the edge of a high platform to avoid damage caused by the robot falling from a height. When using, place the robot on a flat surface. **Placement Safety**: Do not place the robot near the edge of a high platform to avoid damage from falls. Always operate the robot on a flat, stable surface.

2) **Movement Deviation**: If the robot veers significantly to one side and cannot move in a straight line, please refer to 9. IMU, Linear Velocity and Angular Velocity Calibration. Calibration helps reduce deviation, but some hardware-level deviation may still remain. Adjust based on your specific accuracy requirements.

3) **USB Receiver Check**: Before powering on, ensure that the USB handle receiver is properly connected. This receiver is pre-installed in the robot before delivery.

4) **Handle Control Precaution**: Due to the intercommunication feature of the handle control, we recommend disabling this function when multiple robots are operating in the same area to prevent misconnection or control interference.

5) **Lidar Guarding Limitation**: The Ackermann chassis version does not support the “Lidar Guarding” feature.

6) **Startup Mapping Tip**: During initial mapping, position the robot facing a flat wall or a closed cardboard box to allow the lidar sensor to scan more reference points.

7) **Map Integrity**: To ensure a complete map, allow the lidar to scan all 360° around the robot in the area where it will operate.

8) **Large-Scale Mapping**: In larger environments, it is recommended to let the robot first complete mapping of the main area, then proceed to scan finer details of the surroundings.